

FINA6229
Machine Learning in Finance

Project 4
Due Date: 12:00 PM, 9 April 2024
Classify Country Risk

Instruction: You are encouraged to work in groups (but not across groups). However, you should independently write up and submit your own assignments through the CUHK Blackboard under the folder: *FINA6229FA>>Course Contents>>Lecture 4>>Project 4*. Demo code (as an image file) is provided under the same folder. In addition to answer the questions below, you are asked to **add some comments right after the “#” whenever it appears in the demo code**, to explain the purpose of the corresponding code.

As an example,

```
#  
monthly = pd.read_csv('monthly2020.csv', index_col=0, header=None)  
T = monthly.shape[0]
```

you are supposed to submit the code with some comments as follows:

```
# import monthly data using the pandas module  
monthly = pd.read_csv('monthly2020.csv', index_col=0, header=None)  
T = monthly.shape[0]
```

This case introduces you to unsupervised learning and lets you apply your Python knowledge with a simple example. The objective is to cluster countries according to their risk for foreign investment. The following data has been collected for 122 countries:

- GDP growth rate (IMF), in percentage
- Corruption index (Transparency international), 0 (highly corrupt) to 100 (no corruption)
- Peace index (Institute for Economics and Peace), 1 (very peaceful) to 5 (not at all peaceful)
- Legal Risk Index (Property Rights Association), 0 to 10 (with high values being favorable)

The data is in *countryriskdata.csv*.

1. Examine how closely the features are correlated with scatter plots.
2. In the demo code, the corruption index has the most weight because it ranges from 0 to 100. To give the features equal weights, they should be normalized first.

3. Test the effect on inertia (i.e., the within-cluster sum of squares) of increasing the number of clusters. The elbow method argues that the optimal number of clusters is the value of K for the which the marginal gain of adding another cluster drops significantly. Use this criterion to determine the optimal number of clusters. Report the centers for each cluster and the countries in each cluster. Rank the clusters according to country risk.

Lastly, please submit this assignment electronically through the CUHK Blackboard as **a single Jupyter Notebook (*.ipynb) with all the outputs and the answers** to the questions above and the code outputs. **Please name the submitted file with project number, your full name, student ID, and group number as:**

ProjectNumber_StudentName_StudentID_GroupNumber.ipynb,

for example: *Project1_GangLi_12345678_Group1.ipynb.*